

Lesson 10 — Design Challenge — Interactive Alarm

Matches LEGO Teacher Portal lesson: "Putt and Play".

Objective (I can...): I can design a model that responds to at least two events I choose. **CS concept:** Applying events to a purposeful build · **Builds on:** Lessons 6–9. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min): A security camera with AI reacts to two events — motion AND a recognized face. *First-principles question:* "Should both events be needed, or just one, to sound your alarm?"

Materials: LEGO CS & AI kit + device per pair, plan sheets, trigger objects. **Model already built** (Events model from Lesson 6) — the 30 min is plan + code + test, small reconfigs only.

60-minute flow - Warm-up (8): Rapid review — pairs list every event type used (press, tap, distance, color). - **Concept + demo (10):** Challenge: build an "alarm" or "greeter" that reacts to two triggers. Coach models a pseudocode plan for one trigger, then steps back. - **Build/Code (30):** Pairs plan and code two event-driven responses on the built model (motion near → alarm sound; tap → reset light). *Swap at 15 min.* - **Challenge (7):** Add a third trigger or make one response multi-step. - **Share + pack-up (5): Gadget fair — "guess my program"** — before a pair triggers their gadget, the audience guesses which events it responds to, then the pair triggers them live to check. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator checks plan; swap at 15 min. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a one-event target with a partial plan to finish. *Extension / fast-finisher* — require distinct sensor + control triggers. *Reteach* — reuse a Lesson 7–9 program as a starting template.

Exit check: Pair delivers a working model with two+ distinct event responses and a matching plan.

Progress layer: Send home the **Unit 2 parent note** ("your child can make a model sense and react") with the **build photo**. **At-home (optional):** Invent an alarm for something at home; list its triggers. ~10 min.